Modelling with Differentiation (Optimisation)

Exam code: 9MA0

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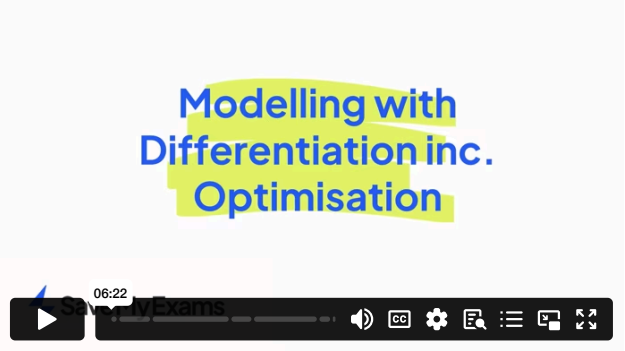

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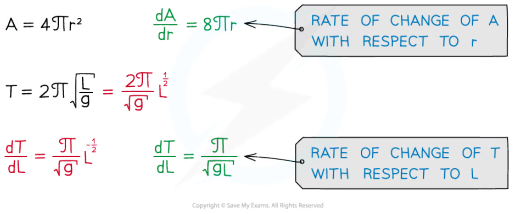
Modelling with differentiation (optimisation)

How do I find the maximum or minimum values of models?

- Derivatives can be calculated for any variables – not just ***y*** and ***x***
- In every case the derivative is a formula giving the rate of change of one variable with respect to the other variable



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

minimisation problems in modelling questions

e.g. AT TIME t SECONDS THE HEIGHT OF A BALL ABOVE THE GROUND IN METRES IS GIVEN BY FORMULA

$$h = 20t - 5t^2$$

USE DIFFERENTIATION TO FIND THE MAXIMUM HEIGHT REACHED BY THE BALL.

$$\frac{dh}{dt} = 20 - 10t$$

$$20 - 10t = 0$$

$$t = 2$$

THE GRAPH OF $20t - 5t^2$ IS A "DOWN" PARABOLA. THE MAXIMUM VALUE OCCURS WHEN $\frac{dh}{dt} = 0$

WHEN $t = 2$,

$$h = 20(2) - 5(2)^2 = 20 \text{ METRES}$$

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Examiner Tips and Tricks

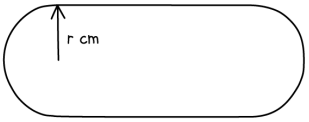
Exam questions on this topic will often be divided into two parts:

- First a 'Show that...' part where you derive a given formula from the information in the question
- And then a 'Find...' part where you use differentiation to answer a question about the formula

Even if you can't answer the first part you can still use the formula to answer the second part.

Worked Example

with a semicircle on each end, as shown in the diagram below.



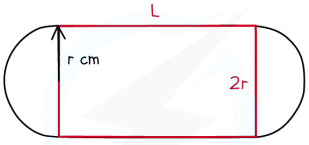
The total area of the bed is to be 100π m^2.

a) Show that the perimeter of the bed is given by the formula $P = \pi\left(r + \frac{100}{r}\right)$

b) For what value of r is the rate of change of P with respect to r equal to $-\frac{9\pi}{16}$ metres per metre?

c) Find the perimeter of the bed for the value of r found in part (b).

a)



THE WIDTH OF THE RECTANGLE IS 2r AND THE LENGTH IS L. THE AREA OF THE BED IS

$$A = \frac{1}{2}\pi r^2 + 2rL + \frac{1}{2}\pi r^2 = \pi r^2 + 2rL$$

SEMICIRCLE RECTANGLE SEMICIRCLE

THE AREA IS 100π, SO

$$\pi r^2 + 2rL = 100\pi$$

$$2rL = 100\pi - \pi r^2$$

$$L = \frac{50\pi}{r} - \frac{\pi}{2}r$$

USE GIVEN AREA TO WRITE L IN TERMS OF r

THE PERIMETER OF THE BED IS

$$P = \pi r + \pi r + 2L = 2\pi r + 2L$$

SEMICIRCLES

$$P = 2\pi r + 2\left(\frac{50\pi}{r} - \frac{\pi}{2}r\right)$$

$$= 2\pi r + \frac{100\pi}{r} - \pi r$$

$$= \pi r + \frac{100\pi}{r}$$

$$P = \pi\left(r + \frac{100}{r}\right)$$

SUBSTITUTE VALUE FOR L INTO PERIMETER FORMULA

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